Revisiting Neutral Baselines in Sentence Processing: Evidence from Self-Paced Reading and Eye Movements

Classic studies of context effects frequently contrasted predictable, unpredictable, and neutral or no-context conditions to estimate facilitation and inhibition during word recognition (e.g., Stanovich & West, 1981). Importantly, Stanovich and West (1988) directly evaluated several candidate baseline frames in a series of empirical studies and reported that neutral, low-constraint contexts, most notably the sentence frame "*The next word will be…*", elicited the fastest response times relative to other baseline frames (e.g., "*They said it was the..*", randomized word strings, and strings of Xs). On this basis, frames as "*The next word will be…"* were argued to function as natural control conditions against which facilitation and inhibition effects could be interpreted. Here, we directly test this assumption using two experiments that provide moment-to-moment online measures of sentence processing, self-paced reading and eye-tracking during reading, each contrasting predictable and unpredictable words embedded within the same sentence frame against a neutral sentence frame. Based on Stanovich and West (1988), we hypothesized that the neutral condition would pattern intermediate between predictable and unpredictable sentence continuations.

Experiment 1. In the self-paced reading (SPR) experiment (N = 67 native English speakers), we compared predictable and unpredictable words embedded within the same sentence frame ("He scraped the cold food from his plate/spoon before washing it") against the same critical words embedded in a neutral sentence frame (e.g., "*The next phrase will be plate/spoon before washing it").* Region RTs were computed as the sum of the target and its two-word spillover region. A linear mixed-effects model revealed a robust main effect of Condition, $F(2, 1733) = 29.00$, $p < .001$. Predictable completions yielded the fastest reading (emmean = 1042 ms), Unpredictable completions were slower (1101 ms), and Neutral items elicited the slowest reading times (1173 ms) (Figure 1). All pairwise comparisons were significant. Notably, the Neutral disadvantage was as large, or larger, than the regular predictability effect (Unpredictable > Predictable), suggesting that contexts lacking constraint impose additional processing effort rather than functioning as a low-effort baseline.

Experiment 2. A separate eye-tracking study (N=46 native English speakers) replicated and extended these findings using three fixation-based measures: gaze duration (GZD), regression path duration (RPD), and total time (TT). After standard preprocessing (merging short fixations; exclusion of isolated fixations <80 ms; 4-MAD filtering), linear mixed-effects models on log-transformed measures revealed significant effects of Condition for GZD, $F(2, 157.16) = 24.36$, $p < .001$; RPD, $F(2, 146.32) = 30.79$, $p < .001$; and TT, $F(2, 167.69) = 99.35$, $p < .001$. Across all measures, predictable words were read fastest (e.g., GZD = 205 ms; TT = 283 ms), unpredictable words were intermediate (GZD = 226 ms; TT = 347 ms), and neutral words were slowest by a wide margin (GZD = 255 ms; TT = 562 ms) (Figure 2). Bonferroni-corrected pairwise comparisons confirmed reliable differences across all contrasts (Predictable < Unpredictable < Neutral), with particularly large effects for total time (predictable vs. neutral: $d = -1.38$) (Figure 2c). Thus, both early and late eye-movement measures converge on the same pattern observed in the SPR task.

Across two methods, the predictable condition robustly facilitated reading, the unpredictable condition incurred moderate processing costs, and the neutral condition produced the largest slowdowns. These results diverge from our initial hypothesis and challenge the assumption that neutral sentences provide cognitively minimal baselines. Rather, they suggest that unconstraining contexts may increase representational uncertainty and processing effort, consistent with accounts emphasizing uncertainty reduction in online comprehension (Levy, 2008). Together, the findings highlight the need to reconsider the use and interpretation of "neutral" baselines in sentence-processing research.

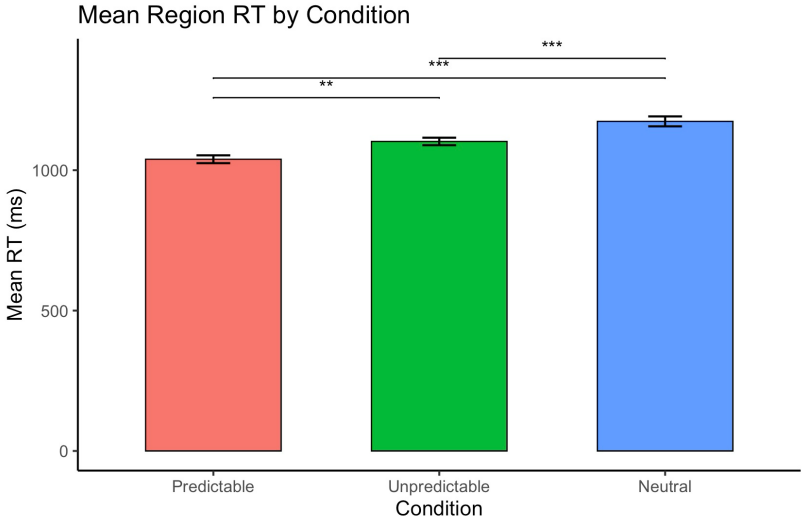

Figure 1. **Experiment 1** (SPR study). Mean summed reading times (ms) for the three-word spillover region as a function of condition (Predictable, Unpredictable, Neutral). Error bars represent ±1 standard error of the mean. Asterisks denote Bonferroni-corrected significance.

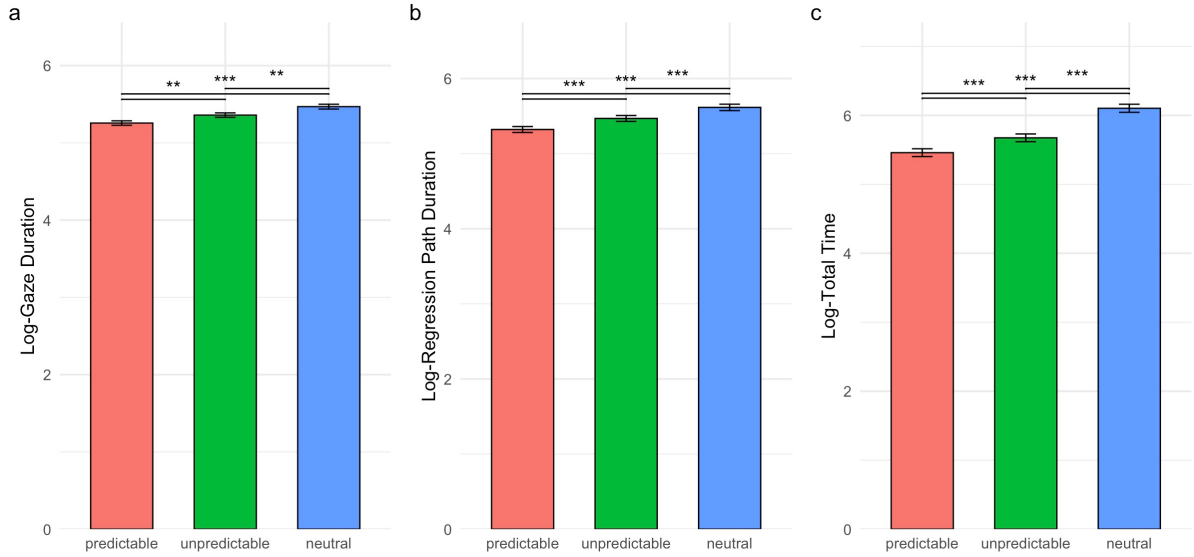

Figure 2. **Experiment 2** (Eye-tracking study). Estimated marginal means of log-transformed fixation durations by predictability condition for (a) Gaze Duration, (b) Regression Path Duration, and (c) Total Time. Error bars represent ±1 standard error of the mean. Asterisks denote Bonferroni-corrected significance.

References

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Stanovich, K. E., & West, R. F. (1981). The effect of sentence context on ongoing word recognition: Tests of a two-process theory. *Journal of Experimental Psychology: Human Perception and Performance*, 7(3), 658–672.
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